

CODING CHALLENGE – 04

Question:

You are given three Excel tables:

1. Sales (OrderID, OrderDate, ProductID, Quantity, SalesAmount)
2. Products (ProductID, ProductName, Category)
3. Regions (RegionID, RegionName)

Management asks for an interactive dashboard that shows:

- Monthly sales trend and weekday-wise performance
- Top 5 products by sales
- Sales contribution by category and region
- Automatic insights and quick summaries for decision-making

Explain how you would:

1. Use date & time functions to derive meaningful time-based insights
2. Connect multiple tables using lookup/reference functions or Pivot Table relationships
3. Build Pivot Tables and Pivot Charts for analysis and visualization
4. Create a dynamic, interactive dashboard using slicers and filters
5. Leverage Excel AI features (Quick Analysis / AI analyst) to generate insights efficiently

Justify why you choose specific functions, pivot features, and chart types in your approach.

ANSWERS :

1. Using Date & Time Functions for Time-Based Insights Steps

- Add helper columns in the Sales table:

Month =TEXT(OrderDate,"mmm-yyyy")

Month Number (for correct sorting) =MONTH(OrderDate)

Weekday Name =TEXT(OrderDate,"dddd")

Weekday Number =WEEKDAY(OrderDate,2)

MONTH() and TEXT() allow grouping sales monthly instead of daily noise.

WEEKDAY() helps analyze weekday-wise performance (business vs weekend behavior).

Month and weekday numbers ensure logical sorting in Pivot Tables and charts.

2. Connecting Multiple Tables (Sales, Products, Regions)

Data Model

- Load all three tables into Excel Data Model.
- Create relationships:
Sales[ProductID] → Products[ProductID]
Sales[RegionID] → Regions[RegionID]

3. Building Pivot Tables and Pivot Charts Monthly Sales Trend

- Pivot Table:
Rows: Month
Values: Sum of SalesAmount
- Pivot Chart: Line Chart

Justification

Line charts clearly show trends and seasonality over time.

Top 5 Products by Sales

- Pivot Table:
Rows: ProductName
Values: Sum of SalesAmount
Apply Top 5 Value Filter
- Pivot Chart: Bar Chart

Justification

- Bar charts highlight ranking clearly.
- Top-N filtering focuses attention on key performers.

Sales Contribution by Category and Region

- Pivot Table:

Rows: Category / Region

Values: Sum of SalesAmount

- Pivot Chart:

Category: Donut or Pie

Region: Stacked Column or Bar

Justification

- Donut/Pie charts show proportional contribution.
- Stacked charts compare contribution across multiple regions or categories.

4. Creating a Dynamic, Interactive Dashboard

- Place all Pivot Charts on a single dashboard sheet.
- Insert Slicers for:

Month

Category

Region

Product

- Connect slicers to all relevant Pivot Tables.

5. Leveraging Excel AI Features for Insights

Quick Analysis

- Select Pivot output or summarized data.
- Use Quick Analysis → Charts / Totals / Trends.

Analyze Data (AI Analyst)

- Ask questions like:

“Which product has the highest sales growth?”

“Which weekday performs best in each region?”

Why Excel AI features

- Saves time in exploratory analysis.
- Helps generate automatic insights and summaries.
- Useful for management discussions without manual calculations.